

AI-Powered Fake News Detection System Using Machine Learning

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SDG Alignment: SDG 11 – Sustainable Cities and Communities

Prepared By: [Y. Maria Jency]

College Name: [Kamaraj College]

1M1B AI for Sustainability Virtual Internship

Project Overview

Project Description

The rapid growth of digital media and social networking platforms has increased the spread of fake news and misinformation. False information can influence public opinion, create panic, and negatively impact society. This project develops an AI-powered system that automatically detects fake news using Machine Learning techniques and helps users verify information before sharing it.

SDG Alignment

Sustainable Development Goal (SDG 11)

Sustainable Cities and Communities

How This Project Supports SDG 11

- Promotes access to reliable information.
- Reduces misinformation and public confusion.

- Encourages responsible digital communication.
 - Supports informed decision-making.
 - Strengthens community resilience and trust.
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Problem Statement

The widespread use of social media and online news platforms has made it easy for fake news to spread rapidly. Misinformation can create social unrest, influence opinions, and cause panic during critical events such as elections, natural disasters, and health emergencies. Manual fact-checking is time-consuming and difficult due to the enormous volume of content generated daily. Therefore, an automated AI-based solution is required to identify and classify fake news accurately.

Proposed Solution

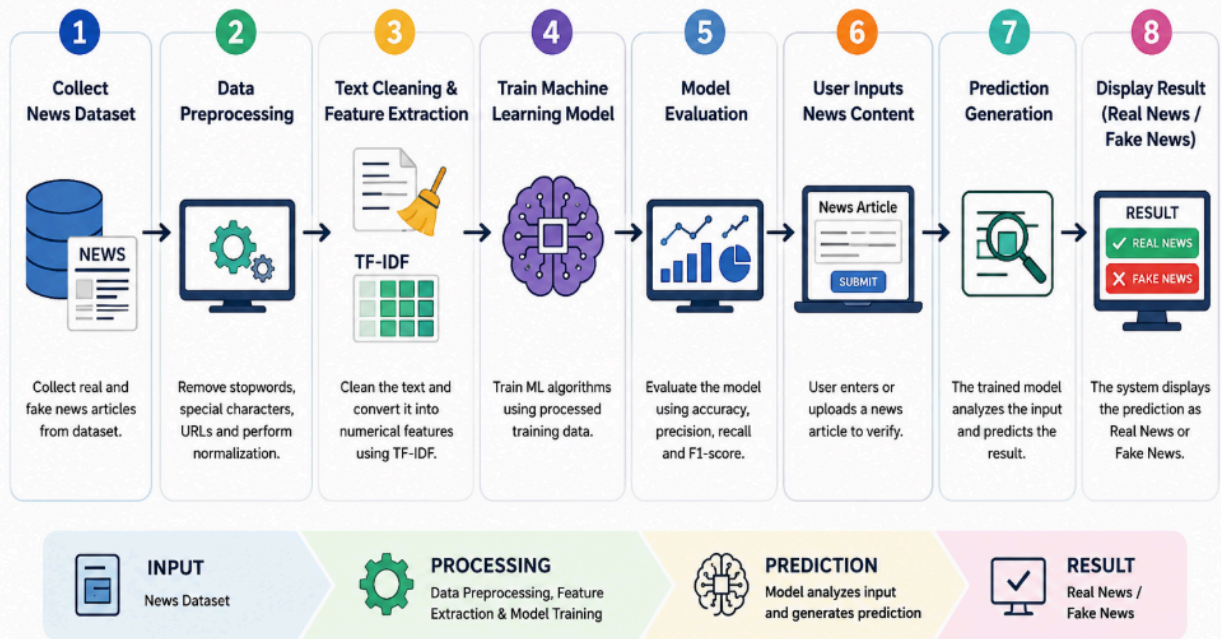
The proposed solution uses Machine Learning and Natural Language Processing (NLP) techniques to analyze news articles and determine their authenticity.

Key Features

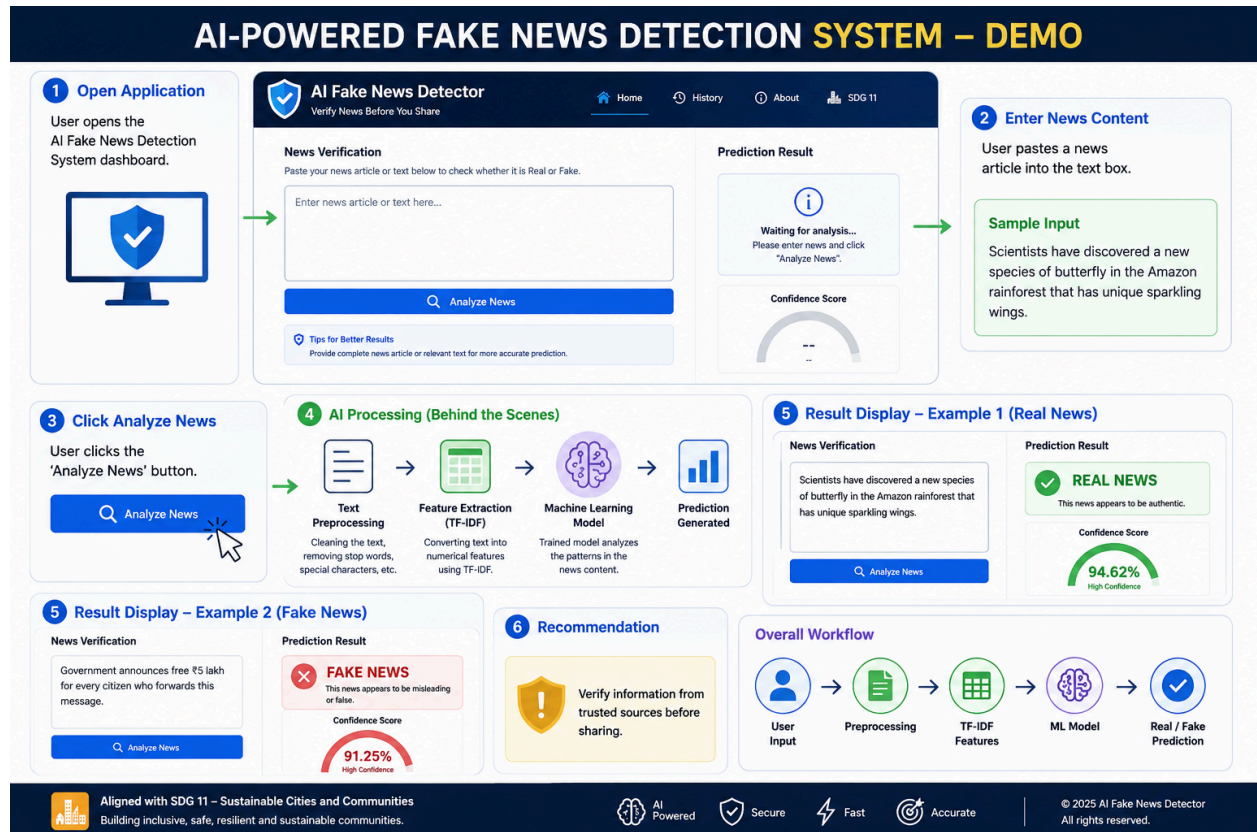
- Automatic fake news detection.
 - Instant prediction results.
 - User-friendly interface.
 - Machine Learning-based classification.
 - Supports reliable information sharing.
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System Architecture / Workflow

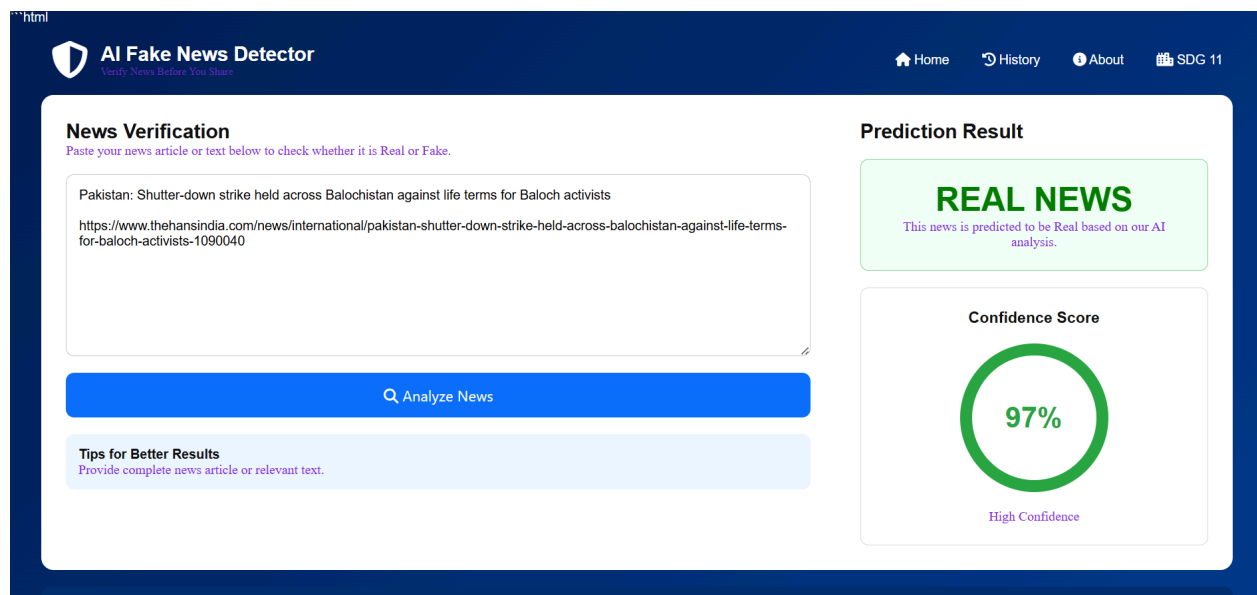
WORKFLOW



Prototype(DEMO)

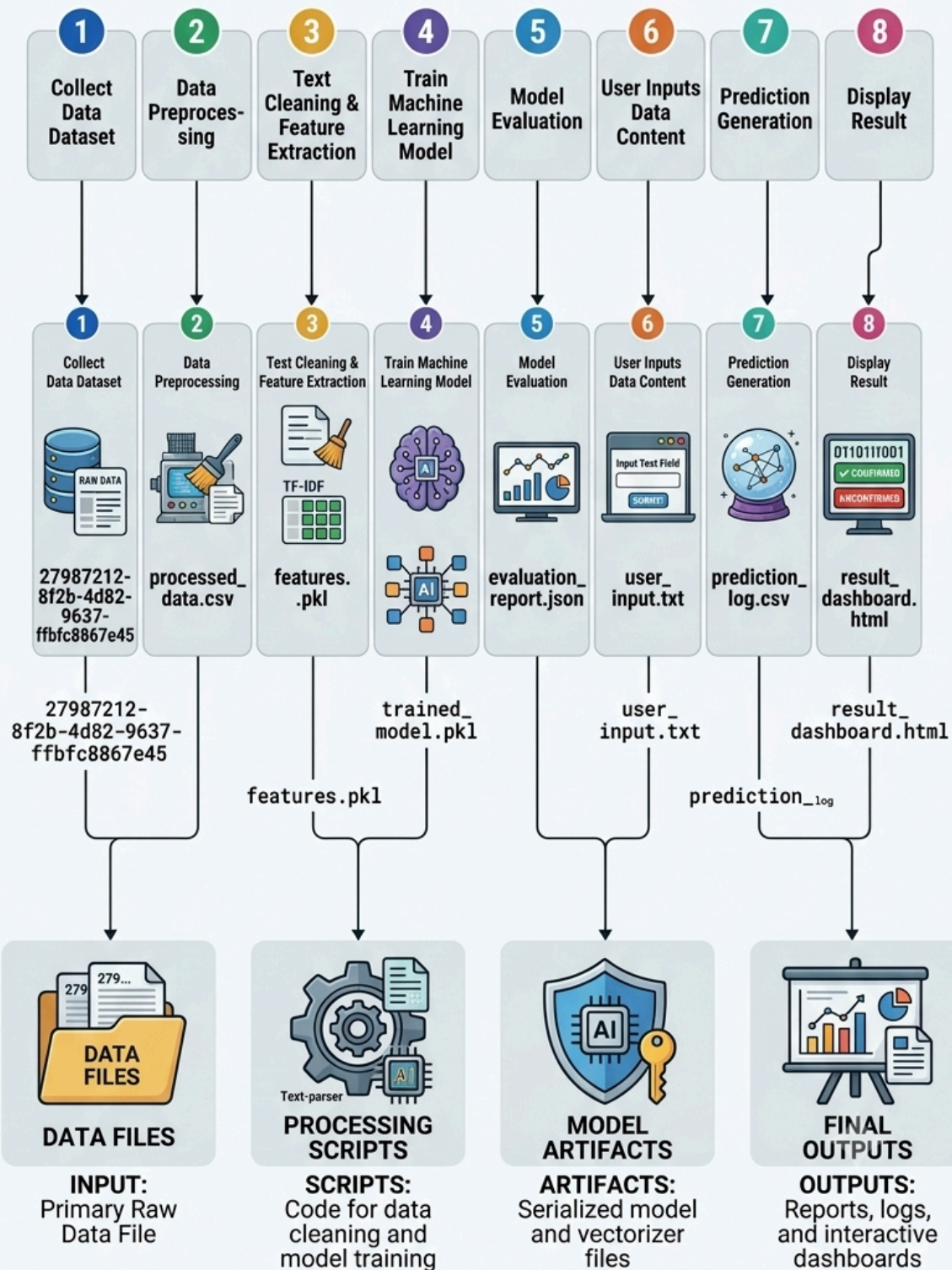


Frontend output



Project Structure

DATA SCIENCE PROJECT WORKFLOW & FILE STRUCTURE



Dataset Description

Dataset Information

Dataset Name: Fake and Real News Dataset

Source: Kaggle

Dataset Statistics

- Total Articles: 44,919
- Fake News: 23,502
- Real News: 21,417

Features

Feature	Description
Title	News Headline
Text	Article Content
Subject	News Category
Date	Publication Date

Technologies Used

Technologies and Tools

Programming Language

- Python

Libraries

- Pandas
- NumPy
- Scikit-learn
- NLTK
- Matplotlib

Machine Learning Algorithms

- Logistic Regression

- Naïve Bayes
- Random Forest

Frontend

- HTML
- CSS
- JavaScript

Development Environment

- Jupyter Notebook
 - VS Code
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Objectives

- Detect fake news automatically.
 - Improve information reliability.
 - Reduce misinformation spread.
 - Enhance digital literacy.
 - Support responsible information sharing.
 - Promote informed decision-making.
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Target Users

Who Benefits From This System?

- Social Media Users
 - Students
 - Researchers
 - Journalists
 - Educational Institutions
 - Government Agencies
 - General Public
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Responsible AI Considerations

Responsible AI Practices

- Fair and unbiased predictions.
 - Privacy and data protection.
 - Transparent decision-making.
 - Reliable dataset usage.
 - Regular model updates.
 - Ethical use of Artificial Intelligence.
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Expected Impact & Conclusion

- Reduces the spread of misinformation.
- Increases public awareness.
- Improves trust in online information.
- Encourages responsible digital behavior.
- Creates safer and more informed communities.

Conclusion

The AI-Powered Fake News Detection System provides an effective solution for identifying misinformation using Machine Learning. By helping users verify news content quickly and accurately, the system promotes trustworthy information sharing and contributes to Sustainable Development Goal 11 by building informed, resilient, and sustainable communities.

Thank You